

Bottleneck-Guided Keccak Acceleration on an RV32 SIMT GPU

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Abstract—We study bottleneck selection and instruction granularity for post-quantum cryptography on an RV32 Vortex SIMT GPU. Software ablation attributes 65.65% of an instrumented ML-KEM-768 round trip to Keccak-[1600] and 12.67% to forward and inverse number-theoretic transforms, giving elimination bounds of 2.911 and 1.145. We retain C and RV32 assembly baselines, then implement THETA, RHOPI, and CHII/IOTA as six single-destination instructions. Twenty-five lanes hold a state in ordinary registers. A whole-round register instruction pair and a memory-pointer permutation engine provide comparison points. On the same 8-warp, 32-thread processor, stage instructions improve RTL permutation throughput by 12.91 times over shuffle-only SG25. With the final cooperative NTT and arithmetic path fixed, the stage backend improves complete ML-KEM cycles by 5.128 times at one request and 3.660 times at eight requests over serial software; the instruction-only gains over the identical shuffle mapping are 1.460 and 1.566 times. At a common 250 MHz target, the stage extension increases full-core LUTs by 1.22% and flip-flops by 0.73%. Whole-round fusion raises permutation throughput by another 4.093 times, but improves the final integrated ML-KEM by only 1.041 and 1.025 times. A matched integer-only RV64 port halves the Keccak collective issue count. At application scope it reduces Stage/KROUND instructions by 7.00%/5.73% and one-request cycles by 4.07%/2.98%, while eight-request cycles rise slightly. Collective granularity trades pipeline storage against benefits that shrink substantially at application scope.

Index Terms—post-quantum cryptography, ML-KEM, ML-DSA, Keccak, RISC-V, SIMT, instruction-set extension, FPGA

I. INTRODUCTION

ML-KEM and ML-DSA standardize module-lattice mechanisms for key establishment and digital signatures [1], [2]. Their implementations combine polynomial arithmetic with SHA-3 and SHAKE [3]. A processor designer can accelerate an NTT butterfly, a Keccak stage, or an entire permutation, but the most parallel primitive need not provide the largest application benefit. The baseline software, the distribution of work among lanes, and the fraction of time outside the accelerated primitive all influence the result.

An extensible GPU makes this choice particularly consequential. Vortex supplies an open RISC-V SIMT hardware and software stack [4], allowing custom instructions to be evaluated through software, functional simulation, processor RTL, and synthesis. A pointer instruction can send a memory-resident Keccak state to a dedicated processing element (PE). Alternatively, a subgroup can hold the state in ordinary lane registers and cooperate through reductions and fixed gathers. Both may be fast, but they exercise different parts of the GPU and impose different integration costs.

This work asks three questions. First, how much of an unaccelerated PQC request is actually attributable to Keccak and NTT, with a stated input and denominator? Second, how much benefit comes from a cooperative software mapping before adding an instruction? Third, how do stage-level and whole-round collectives compare with a pointer PE after accounting for complete requests and full-core implementation cost?

Our contributions are as follows.

- 1) We assemble a reproducible baseline study covering reference C, optimized RV32 Keccak assembly, request batching, cooperative software NTT, and component ablation. The study distinguishes measured cycle differences from extrapolated or algebraic quantities.
- 2) We implement a stage-factored subgroup interface with ordinary GPR state, explicit RV32 low/high destinations, an RV64 full-word port, and a full-warp contract. All eight combinations of software and instruction stages are evaluated on the same hardware.
- 3) We implement a whole-round GPR comparator and evaluate it alongside the stage design and pointer PE. The comparison connects instruction granularity to RTL throughput, ML-KEM latency, memory-operation counts, FPGA full-core area/timing, and isolated ASIC cell area.
- 4) We select a register-resident W32 NTT with a shared 16-multiplier bank, validate its complete-KEM behavior, and hold this implementation fixed in the final Keccak performance and full-core area comparison. A matched RV64 portability experiment measures both complete-KEM behavior and implementation cost without changing the narrow NTT arithmetic.

We do not claim the 25-thread state layout or whole-round warp collectives as new concepts. ML-Cube uses 25-thread Keccak with shuffles [5], and an NVIDIA patent application describes a 64-bit warp-collective Keccak round [6]. Our focus is the explicit RV32/RV64 interface, its open implementation, and the controlled comparison of granularities. The measured results do not select one universal winner: the whole-round unit leads the GPR permutation test, while the pointer PE leads the eight-request ML-KEM test.

II. BACKGROUND AND EXPERIMENTAL SCOPE

A. PQC work and two forms of parallelism

The measured ML-KEM-768 request includes key generation, encapsulation, and decapsulation; the ML-DSA-65 round

trip includes key generation, signing, and verification. The software is based on `mlkem-native` and `mldsa-native` [7], [8]. We call the number of concurrent requests M . Increasing M offers independent work to the warp scheduler without shortening a particular request’s dependency chain.

We use L for software lanes assigned to work within a request. In the historical x4 backend, these lanes process independent hash streams exposed by the library API. In SG25, lanes cooperate on *one* permutation: lane $t = x + 5y$ holds the 64-bit word $A[x, y]$, for $0 \leq x, y < 5$. These mappings have different synchronization and storage requirements. Neither their speedups nor their resource counts can be multiplied together without an experiment that combines them.

A Keccak-f[1600] permutation consists of 24 rounds. Each round computes column parity and the THETA correction, rotates and rearranges words in RHOPI, then applies CHI and the IOTA round constant. This produces structured cross-lane communication: five-word column reductions, a fixed permutation, and three-word row gathers. The RV32 interface stores each word in two 32-bit registers, avoiding a change to the core-wide integer width.

B. Measurement families and denominators

Table I separates the measurement families. All cycle results are simulation results. FPGA numbers come from routed out-of-context `VX_core_top` or isolated-unit implementations targeting the Alveo V80 part; they are not board measurements. We preserve historical software experiments to explain the design choice, but use the 8-warp, 32-thread experiments for the collective comparison.

The principal collective configuration has a 32-lane ALU and LSU, 16 KiB instruction and data caches, and 16 KiB local memory. L2/L3 are disabled. Historical cache sweeps explicitly vary L2; their numbers are not merged with the main configuration. SimX provides functional and timing models, processor RTL simulation provides cycle comparisons, and XRT simulation checks integration through the accelerator interface.

For an equal-sized batch, speedup is

$$S_{\text{same } M} = \frac{C_{\text{baseline}}(M)}{C_{\text{candidate}}(M)}. \quad (1)$$

For scaling a backend from one request to M , throughput gain is

$$G_M = \frac{MC(1)}{C(M)}. \quad (2)$$

These are different denominators. Whole-launch cycles include device-side launch setup and teardown; they exclude host transfers. Phase cycles are intervals around the cryptographic calls. A phase total cannot silently replace a whole-launch denominator.

III. SOFTWARE BASELINES AND BOTTLENECK ATTRIBUTION

A. ML-KEM: Keccak, transforms, and the residual

The historical ML-KEM profile uses the same FIPS 203 known-answer-test coins as its corresponding baseline. It

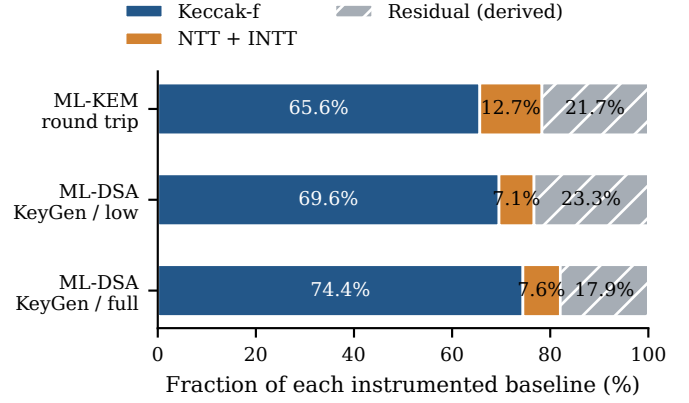


Fig. 1. Historical component attribution from separate ablations. Each bar has its own instrumented denominator. Hatched residuals are obtained by subtraction; additivity has not been measured for these rows.

executes 144 scalar permutations, including the scalar calls made through x4 fallback, and 15 forward plus nine inverse transforms. The x4 counter records 24 wrapper calls; *it is not added as another 4×24 permutations* to the scalar count.

We remove either the permutation body or the forward/inverse transform bodies through native hooks. The cryptographic outputs of these ablated programs are intentionally invalid. They are used only to estimate component cost, with the instrumented baseline as the denominator:

$$f_i = \frac{C_0 - C_{-i}}{C_0}, \quad S_{\infty, i} = \frac{C_0}{C_{-i}} = \frac{1}{1 - f_i}. \quad (3)$$

The removed bodies retain the surrounding invocation path. The cycle differences therefore include consequences for scheduling and memory, rather than being a count of executed ALU operations alone.

Table II gives 21,986,475 removed cycles for Keccak and 4,243,182 for NTT/INTT. Relative to 33,492,153 profile cycles, the shares are 65.65% and 12.67%. Thus even ideal transform elimination yields only $1.145\times$ on this workload, whereas ideal Keccak elimination yields $2.911\times$. The unchanged permutation count across arms rules out a change in the number of permutations as the explanation for the Keccak delta. It does not establish that every data-dependent path is invariant.

Subtracting both deltas from the baseline leaves 7,262,496 cycles, or 21.68%. Figure 1 shows this *algebraic residual*. It is not a separately instrumented exclusive category or a measured joint ablation. It can contain sampling, polynomial operations outside NTT/INTT, packing, decoding, absorb/squeeze work, allocation, and control. The archive does not split these contributors further, so we do not label all residual time as memory or claim a measured combined Keccak-plus-NTT bound.

Input consistency matters because rejection sampling can change the permutation count. The archived profile now uses KAT coins; earlier 156-permutation measurements and stale narrative percentages are excluded from this attribution. The

TABLE I

EXPERIMENTAL FAMILIES. ALL USE ONE CORE AND RV32 EXCEPT THE MATCHED RV32/RV64 PORTABILITY ROWS. W IS RESIDENT WARP CAPACITY, T IS HARDWARE THREADS PER WARP. A REQUEST COUNT IS A SEPARATE QUANTITY.

Family	$W \times T$	Workload / software	Timing denominator	Scope of valid comparison
Historical attribution	4×4	Single-lane ML-KEM KAT; ML-DSA fixed input	Instrumented profile's own cycles	Component ablation within that profile
C / assembly A/B	4×4	Same ML-KEM request; reference C or PQRV	Whole-launch device cycles	Optimized software versus C
Batch and x4 mapping	8×4	$M = 1, 2, 4, 8$; $L = 1, 2, 4$ where archived	Each sweep's own $M = 1, L = 1$ launch	Request throughput and cache/mapping interaction
SG25 stages / round	8×32	32-wide SIMD/ALU; 1 or 8 states	Synchronized permutation span	Shuffle, stage-use ablation, whole round
Complete SG25 ML-KEM	8×32	Serial FIPS-202; 143 permutations/request	Whole-launch cycles at fixed M	SG1, shuffle, stages, whole round, pointer
Final NTT+Keccak ML-KEM	8×32	W32 NTT/arithmetic; KAT; 140 permutations/request	Whole-launch cycles at fixed M	Five Keccak backends with all other arms fixed
Physical implementation	8×32	RV32 backend builds; matched integer-only RV32/RV64 NTT/Stage/KROUND builds; isolated units	Matched 250/300 MHz constraints	LUT/FF and STA-derived frequency within each family

TABLE II

ML-KEM-768 COMPONENT ABLATION. SINGLE LANE, $4w \times 4t$, SIMX; EVERY ROW RECORDS 144 PERMUTATIONS.

Variant	Cycles	Removed cycles	Share	Bound
Profile baseline	33,492,153	—	—	—
Keccak removed	11,505,678	21,986,475	65.65%	2.911
NTT/INTT removed	29,248,971	4,243,182	12.67%	1.145

Source: pqc/results/ablation_mlkem.csv.

TABLE III

ML-DSA-65 ATTRIBUTION, SINGLE LANE, $4w \times 4t$, SIMX. M: MEASURED COMPONENT ABLATIONS; E: EXTRAPOLATED COMPONENT COSTS. TOTALS ARE MILLIONS OF PROFILE CYCLES.

Scope / budget	Total	Keccak	NTT/INTT	Basis
KeyGen / low	42.022	69.56%	7.13%	M
KeyGen / full	39.121	74.44%	7.62%	M
Round trip / low	189.935	63.34%	14.78%	E
Round trip / full	142.119	62.76%	18.12%	E

Source: pqc/results/ablation_mldsa.csv.

main SG25 integration instead uses fixed bytes $0, \dots, 95$ and the library's serial FIPS-202 path, with 143 permutations in every compared backend. We do not transfer the historical 65.65% share or its bound to that new software/configuration.

B. ML-DSA: measured key generation and estimated round trips

Table III gives the corresponding ML-DSA results under two library memory budgets. Key generation admits the same ablation method: both budgets perform 190 permutations. The Keccak deltas are 29,231,666 cycles for `low` and 29,119,995 for `full`, within 0.4%. The resulting elimination bounds are $3.285\times$ and $3.912\times$. NTT/INTT contribute 7.13% and 7.62%, respectively.

Signing and verification do not admit this substitution. Stubbing the hash during signing changes the rejection loop; stubbing it during verification can make `SampleInBall`

TABLE IV

REFERENCE C VERSUS PQRV ASSEMBLY ON THE SAME HISTORICAL SINGLE-LANE ML-KEM CONFIGURATION. RATIO IS C/ASSEMBLY.

Metric	Reference C	RV32 assembly	Ratio
Permutation (cycles)	151,872.1	134,320.2	1.131
KEM launch (cycles)	33,066,490	30,615,811	1.080
Retired instructions	3,467,083	3,248,627	1.067
Stack peak (bytes)	3,944	3,776	1.044

Source: pqc/results/keccak_baseline_columns.csv.

fail to terminate normally. The round-trip entries therefore multiply call counts by component costs calibrated from key generation. They are estimates, not direct measurements of signing/verification time in each primitive. For the archived input, the low/full round trips execute 782/582 permutations and 103/95 transforms. Their estimated Keccak shares are 63.34%/62.76%; transform shares are 14.78%/18.12%. The memory budget changes the workload composition as well as its total.

These ML-DSA round-trip values describe one input's rejection-loop outcome. An earlier host distribution was not completely archived and had an overflowing histogram; its tail statistics are not used here. A separate, correct pointer-PE experiment reaches $2.652\times/2.628\times$ round-trip speedups for low/full budgets, supporting the scale of the hash-cost estimate without supplying a missing input distribution.

C. An optimized software baseline

Reference C alone is insufficient to evaluate an instruction extension. We therefore also use PQRV's hand-written RV32 Keccak implementation, introduced through the library's native permutation hook [9]. Table IV compares these columns in one historical $4w \times 4t$ experiment. The NTT implementation is unchanged by the swap.

Assembly improves the permutation by $1.131\times$ and the complete launch by $1.080\times$. Using the profile share in

TABLE V
SOFTWARE FORWARD NTT FOR ML-KEM. THE LIBRARY REFERENCE
TAKES 133,526 CYCLES. MISMATCH COUNT IS ZERO IN ALL ROWS.

L	Cooperative cycles	Gain vs library	Gain vs coop. $L = 1$
1	211,506	0.631	1.000
2	106,297	1.256	1.990
4	53,682	2.487	3.940

Source: pqc/results/ntt_cooperative.csv.

$1/(1 - f_K + f_K/s_K)$ predicts approximately $1.082\times$, a cross-check rather than a replacement for the measured launch. In a separately archived, internally matched three-arm run, reference C, assembly, and the pointer PE take 32,778,558, 30,321,833, and 11,527,027 cycles. That run gives a $2.844\times$ PE gain over C and $2.631\times$ over assembly. These counts come from `keccak_ise_simx.csv`; they must not be divided into the totals in Table IV.

D. Cooperative NTT as a control

The cooperative forward NTT maps each layer’s 128 butterflies across L lanes. It uses the library’s arithmetic and twiddle factors and a shared polynomial array, with visibility fences between layers. Every result is checked coefficient by coefficient against the library. Table V reports both the optimized library denominator and the cooperative implementation’s own one-lane denominator.

Four lanes deliver $3.940\times$ scaling of the cooperative code, but only $2.487\times$ over the faster single-lane library. Indexing and fences make the cooperative one-lane arm slower. This distinction prevents a $3.940\times$ scaling result from being presented as the benefit over existing software. The experiment covers a forward transform, not an integrated NTT/INTT extension or complete-KEM speedup.

Microbenchmark setup is another source of bias. The original forward-NTT microbenchmark times coefficient reconstruction with the transform, reporting 160,270.4 cycles rather than the 133,526-cycle transform-only reference. Multiplying such a setup-inclusive cost by application call counts would overstate the transform share. We use application ablation for attribution and transform-only timing for this software control.

E. Request batching, lane width, and memory budget

Table VI shows that scheduling independent requests recovers throughput without an instruction extension. ML-KEM reaches $6.654\times$ at eight requests on an eight-warps, four-thread core. ML-DSA has archived points through four requests in this baseline family. The table reports $MC(1)/C(M)$, not reduced latency of one request.

Increasing request concurrency and within-request lane width together can be counterproductive. Figure 2 preserves an independent historical ML-KEM x4 sweep with 156 scalar-equivalent permutations for its one-lane arm. At $M = 8$, L2 off, increasing L from one to four changes the throughput gain from 6.739 to 4.016. With L2 on, the corresponding values are 7.145 and 8.280. This is evidence for a cache-sensitive

TABLE VI
UNMODIFIED-SOFTWARE REQUEST BATCHING, $8w \times 4t$, ONE ACTIVE
LANE PER REQUEST, L2/L3 OFF. CYCLES ARE MILLIONS PER COMPLETE
BATCH.

Scheme	M	Batch cycles	Throughput gain
ML-KEM	1	33.569	1.000
ML-KEM	2	33.575	2.000
ML-KEM	4	34.058	3.943
ML-KEM	8	40.361	6.654
ML-DSA	1	193.767	1.000
ML-DSA	2	193.794	2.000
ML-DSA	4	196.700	3.940

Source: pqc/results/batch_baseline.csv.

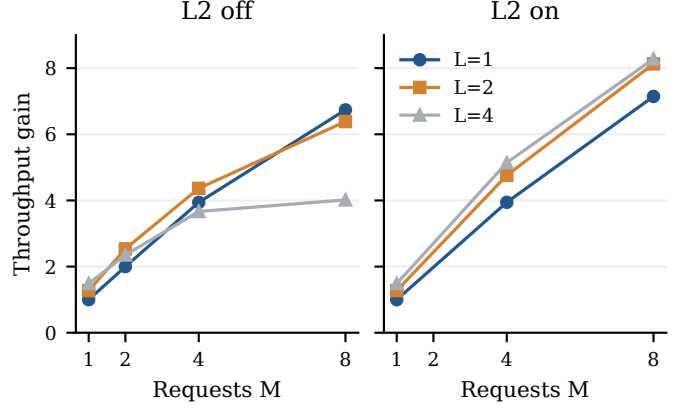


Fig. 2. Historical ML-KEM x4 mapping sweep on $8w \times 4t$. Both panels use the sweep’s L2-off, $M = 1, L = 1$ baseline of 36,614,668 cycles. Source: `mlkem_MxL.csv`. These inputs/backend differ from the KAT ablation and the serial-FIPS-202 SG25 experiment.

interaction between mappings, not a universal benefit of wider launches.

ML-DSA adds another software choice. Table VII varies library memory budget and x4 lane width for one request. The full-budget implementation uses an 86,912-byte arena rather than 21,568 bytes. With the low budget, four lanes provide no gain; with the full budget they improve launch time by $1.601\times$ over that budget’s one-lane arm. Changing both budget and mapping produces $2.118\times$ over low-budget, one-lane software, without new hardware.

IV. A SUBGROUP KECCAK INTERFACE

A. State placement and programming contract

One 32-thread warp holds one state. Lanes 0–24 each hold a 64-bit word as two ordinary RV32 GPR values or one RV64 GPR value. Lanes 25–31 carry no state and return zero. All 32 lanes must execute each collective with the full mask at the same instruction. In particular, guarding the collective with `lane < 25` violates the contract. Guarded loads/stores are allowed if the warp reconverges before the collective.

Each instruction snapshots its complete source vectors and writes one destination vector. It has no pointer operand, memory ordering effect, hidden round counter, or private architectural state. Ordinary RAW hazards, backpressure, and

TABLE VII
ML-DSA SOFTWARE MEMORY-BUDGET/WIDTH CONTROL, $8w \times 4t$, ONE
REQUEST, L2/L3 OFF. GAIN IS RELATIVE TO LOW-BUDGET $L = 1$.

Budget	L	Mcycles	Arena bytes	Gain
low	1	189.927	21,568	1.000
low	2	191.049	21,568	0.994
low	4	191.500	21,568	0.992
full	1	143.574	86,912	1.323
full	2	107.933	86,912	1.760
full	4	89.671	86,912	2.118

Source: pqc/results/mldsa_ram_x_lane.csv.

writeback arbitration still apply. Transient pipeline registers are microarchitectural storage and are included in the area cost.

The implementation supports RV32 and RV64 and requires 32 threads, a 32-wide SIMD datapath, and 32 ALU lanes. A narrower packetized ALU would need operand collection across packets; warp width alone is insufficient. The 25-of-32 state layout uses 78.125% of lanes for state words. The remaining lanes still consume execution resources. We do not assume that this utilization loss is free.

B. Three stages with single-destination instructions

For $t = x + 5y$, let $A[x, y]$ combine the low/high inputs. All coordinates wrap modulo five. THETA computes

$$C[x] = \bigoplus_{y=0}^4 A[x, y], \quad (4)$$

$$T[x, y] = A[x, y] \oplus C[x - 1] \oplus \text{ROL}_{64}(C[x + 1], 1). \quad (5)$$

KTHETA.L/H.SG25 each read both state halves and select one output half. Cross-half carry bits from the one-bit rotation are part of the operation; two independent 32-bit reductions would be incorrect.

RHOPI combines a fixed rotation with the lane permutation:

$$B[y, (2x + 3y) \bmod 5] = \text{ROL}_{64}(T[x, y], r[x, y]). \quad (6)$$

At destination (x_d, y_d) , the source is $x_s = (x_d + 3y_d) \bmod 5$, $y_s = x_d$. KRHOPI.L/H.SG25 select the corresponding result half. Rotation amounts belong to source coordinates, which is essential when implementing the inverse gather.

CHII evaluates each 32-bit half independently:

$$U[x, y] = B[x, y] \oplus (\neg B[x + 1, y] \wedge B[x + 2, y]). \quad (7)$$

Only lane zero then XORs its half of $RC[q]$. Unlike THETA/RHOPI, KCHII.L/H.SG25 read one state half and a uniform round index q , not both state halves. All lanes, including padding lanes, must supply the same index in $0, \dots, 23$.

The six RV32 encodings use CUSTOM0 (0x0b), $\text{funct7}=0x06$, with $\text{funct3}=0 \dots 5$. Each uses two source-vector reads and one destination-vector write. RV64 uses $\text{funct3}=0/2/4$ for full-word THETA/RHOPI/CHII outputs; THETA and RHOPI leave the second source unused, while

CHII reads the uniform round index there. THETA and RHOPI low/high operations must read the same old state:

$$(a_l, a_h) \rightarrow (t_l, t_h) \rightarrow (b_l, b_h) \rightarrow (a'_l, a'_h). \quad (8)$$

Overwriting a_l before the high THETA operation would break this dependency. Two GPRs describe persistent state capacity, not peak register pressure. The all-stage permutation loop uses five GPRs for state, temporaries, round, and loop limit; its disassembly has no load, store, call, or stack frame.

C. ALU integration and the whole-round comparator

Figure 3 contrasts the architectural boundaries. The stage unit sits under `VX_alu_unit` and uses two registered pipeline stages. THETA forms parity then the correction; RHOPI and CHII use the same latency interface. The direct unit latency is two cycles with initiation interval one. These values are not the core's observed dependent instruction latency.

The source code expresses the fixed cross-lane connections directly. It shares the core's operand and result interfaces, but the current implementation does *not* establish physical sharing of the existing general shuffle network. A claim of shuffle-network area reuse would require a different integration and its own synthesis evidence.

The independent whole-round extension uses CUSTOM2 (0x5b). On RV32, $\text{funct3}=0/1$ selects `KROUND.L/H.SG25`; both source vectors contain the old low/high state, and each issue computes one selected half. On RV64, $\text{funct3}=0$ selects `KROUND.SG25`, one source vector contains the complete state word, and one issue computes the complete round. In both forms, funct7 carries a literal round index. The four registered stages are column parity, THETA, RHOPI, and CHII/IOTA. Direct latency is four cycles and initiation interval is one. There is no hidden state retained between issued instructions. The low/high pair recomputes from the same old state; it is not a dual-writeback macro-operation.

Literal round encoding expands 24 rounds into 48 RV32 or 24 RV64 instructions. The stage kernel retains a round loop with six RV32 or three RV64 collectives, an increment, and a branch per round. Consequently, the measured comparison includes both the instruction granularity and its present code generation. It is not an isolated comparison with identical loop-control instructions.

The pointer KECCAKF baseline remains a full-permutation engine connected through the existing SFU/memory path. It supplies a useful dedicated-engine reference, whereas the GPR variants expose lane-vector data directly to the execution unit. The pointer design requires architectural state transfer through the memory interface for each invocation.

D. Full SHAKE and ML-KEM mapping

The subgroup sponge retains state in lane registers across absorb, permutation, and squeeze within a one-shot call. Absorb uses ordinary loads/XORs; squeeze uses ordinary stores. Incremental SHAKE128 saves a word per lane in its context

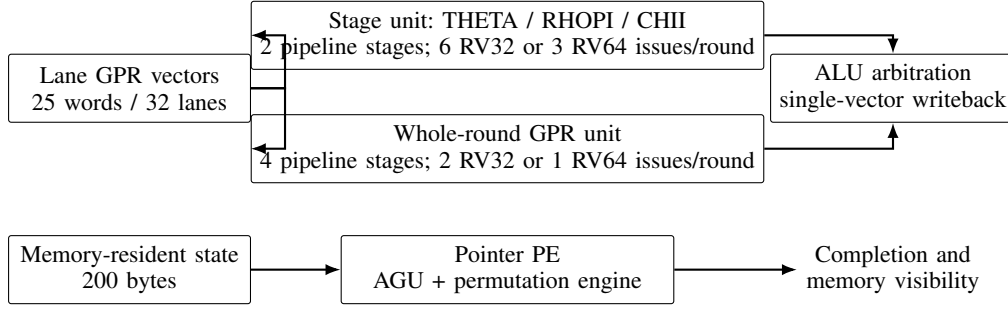


Fig. 3. Alternative Keccak execution boundaries. GPR stages and the whole-round unit use ordinary ALU operand/writeback interfaces. The pointer PE transfers a memory-resident state. Pipeline storage is charged to each implementation.

across separate API calls and reloads it on reentry. Thus register residence within a call does not mean the incremental context never touches memory.

For ML-KEM, lane zero executes the surrounding serial library. A TMC boundary and a noinline worker activate the complete warp for FIPS-202, broadcast arguments, run the subgroup sponge, and restore the leader mask on return. A shared allocation arena avoids allocating the same request’s workspace 32 times.

A correctness-first alternative ran the outer KEM redundantly on every lane. We preserve its results as a control. It satisfies output checks but causes excessive memory operations and contention. Removing that redundancy is a software mapping improvement and must not be attributed solely to the Keccak instructions.

V. EVALUATION

A. Correctness and timing methodology

Stage tests cover aliases, zero registers, high/low bit crossings, padding lanes, all round constants, metadata, and backpressure. The mixed stage RTL unit test has 288 RV32 and 144 RV64 transactions; the whole-round test has 192 RV32 and 96 RV64 transactions. Contract tests diagnose partial masks and invalid rounds, plus nonuniform round operands for CHII. These checks diagnose unsupported programming conditions; they do not define a production trap ABI.

Full-round checks use 17 states and 10,200 word comparisons. Complete SHAKE128/256 tests cover 204 input/output cases, including empty messages, unaligned addresses, and rate boundaries. SimX and XRT integration tests pass. ML-KEM checks return codes, shared-secret agreement, and checksums of the public key, secret key, ciphertext, and shared secret.

Permutation timing launches one state per warp and brackets a dependent chain with synchronized cycle reads. CTA barriers keep initial state loads and final stores outside the timed batch; the measured span is $\max(t_{\text{end}}) - \min(t_{\text{start}})$. It includes scheduling skew and any accesses performed inside software stages. For the eight-arm stage study, the marginal cost is

$$c_{\text{marginal}} = \frac{\text{span}(8) - \text{span}(4)}{4M}. \quad (9)$$

TABLE VIII

RTL MARGINAL PERMUTATION COST FROM EQ. 9. T/R/C INDICATE THETA, RHOPI, CHII INSTRUCTIONS. GAINS USE THE SHUFFLE-ONLY -- ARM AT THE SAME BATCH SIZE.

Arm	1-state cyc.	Gain	8-warp cyc./perm.	Gain
--	19,285.75	1.000	6,215.59	1.000
T-	12,074.00	1.597	3,161.56	1.966
-R-	11,173.25	1.726	3,626.09	1.714
-C	14,760.50	1.307	4,650.69	1.336
TR-	5,275.00	3.656	1,182.75	5.255
T-C	8,818.00	2.187	2,117.38	2.936
-RC	7,708.00	2.502	1,953.38	3.182
TRC	2,050.00	9.408	481.50	12.909

Source: pqc/results/keccak_sg25_stages.csv.

The whole-round comparison uses a separate 64-permutation chain and reports $\text{span}(64)/(64M)$. These estimators are kept separate in the tables.

Across the 32 matched points in the stage dataset, retired instruction counts agree between SimX and processor RTL; maximum whole-launch cycle difference is 3.24%. Passing this set establishes agreement on the tested cases, not a complete proof of the timing model. The complete ML-KEM XRT checks are integration/correctness results; the internally matched SimX rows supply the application comparisons.

B. Eight-way instruction-use ablation

Table VIII uses one hardware build with all six stage encodings enabled. Each arm replaces selected shuffle-software stages with custom instructions. Therefore it measures instruction use, not the physical area obtained by synthesizing only one selected stage.

The complete stage design reduces dependent marginal cost from 19,285.75 to 2,050 cycles, a $9.41\times$ gain. Eight-warp cost falls from 6,215.59 to 481.50 cycles per completed permutation, a $12.91\times$ gain. Figure 4 shows that stage benefits depend on the load point: RHOPI gives the largest single-stage latency improvement, while THETA gives the largest single-stage eight-warp improvement. Combined benefits are not obtained by adding the individual speedups.

C. Whole-round fusion

Table IX gives the longer-chain comparison. The whole-round unit improves RTL single-state cost from 2,052.219 to

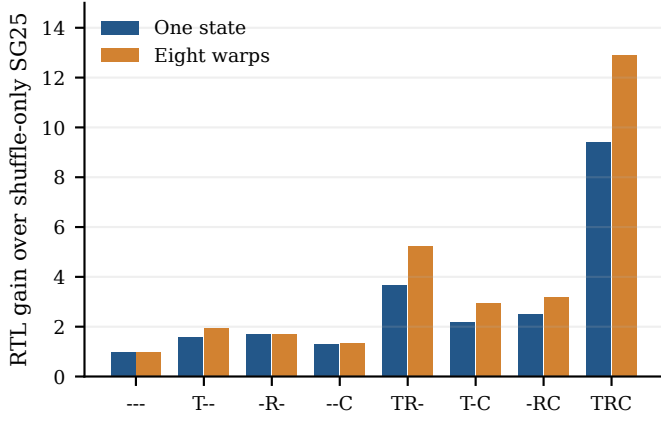


Fig. 4. Stage instruction-use gains on the same $8w \times 32t$ hardware. The latency and throughput denominators differ as defined in Table VIII.

TABLE IX
64-PERMUTATION GPR CHAINS ON $8w \times 32t$. COSTS ARE SPAN/(64M), EXCLUDING INITIAL/FINAL STATE I/O.

Model	Unit	1-state cycles	8-warp cyc./perm.
simx	Stages	2,079.312	479.463
simx	Whole round	502.406	121.727
rtlsim	Stages	2,052.219	481.664
rtlsim	Whole round	499.266	117.693

Source: pqc/results/keccak_kround25.csv.

499.266 cycles and eight-warp cost from 481.664 to 117.693 cycles per completed permutation. This is $4.110\times$ latency improvement and $4.093\times$ throughput improvement.

The result makes whole-round fusion a strong implemented comparator. The reduction from six to two issues is not, by itself, a prediction of these cycle ratios. Loop control, register dependencies, issue arbitration, and pipeline timing contribute to the measured sequence.

D. Complete ML-KEM

Table X compares complete launches on the same core geometry, cache configuration, fixed coins, and serial FIPS-202 path. All arms perform 143 permutations per request. SG1 and shuffle controls run with the stage extension present but unused. The whole-round arm enables its independent extension; the pointer arm enables PQC and KSG25. These enable sets are recorded in the source data.

The shuffle-only cooperative mapping already improves launch cycles by $2.422\times$ at one request and $2.000\times$ at eight requests. Replacing its permutation stages with instructions adds $1.219\times$ and $1.383\times$, yielding total gains of $2.952\times$ and $2.766\times$ over SG1. This matched shuffle denominator isolates the instruction contribution from the substantial benefit of mapping state across registers.

Whole-round fusion adds only $1.019\times$ and $1.013\times$ over the stage backend, despite the roughly fourfold isolated permutation improvement. The single-request instruction count falls from 1,093,915 to 1,073,037, just 1.91%. The evidence identifies a limit to further permutation optimization. It does not yet

TABLE X
COMPLETE ML-KEM-768 ON SIMX, $8w \times 32t$. CYCLES ARE MILLIONS PER LAUNCH; GAINS USE SG1 AT THE SAME M .

Backend	$M = 1$	Gain	$M = 8$	Gain
SG1 C	34.689	1.000	53.176	1.000
SG25 shuffle	14.321	2.422	26.588	2.000
SG25 stages	11.749	2.952	19.228	2.766
SG25 whole round	11.532	3.008	18.989	2.800
Pointer PE	12.430	2.791	17.014	3.126
Redundant outer KEM	28.684	1.209	160.039	0.332

Source: pqc/results/keccak_sg25_mlkem.csv.

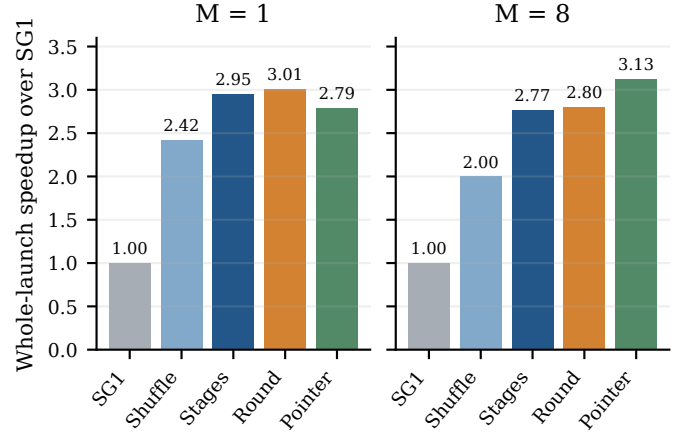


Fig. 5. Complete-launch gains over SG1 at equal request count. The whole-round unit's large permutation advantage produces a small additional application gain. The pointer PE leads the eight-request launch.

quantitatively partition the remainder among absorb/extract, wrappers, transforms, other polynomial work, and allocation.

The stage backend is $1.058\times$ faster than the pointer PE for one request. At eight requests the pointer PE is $1.130\times$ faster than stages and $1.116\times$ faster than the whole-round backend. These results are compatible: a dedicated engine can benefit a saturated request stream even when a register-resident mapping has lower latency for a single request.

E. Phase costs and memory-operation evidence

Table XI makes the single-request phase totals explicit. The final column is launch cycles minus the sum of three measured call intervals; it is not an attributed cryptographic primitive. For SG1, key generation, encapsulation, decapsulation, and this remainder account for approximately 28.36%, 31.82%, 38.39%, and 1.43% of the launch. For stages they account for 23.74%, 32.60%, 39.56%, and 4.10%. These phase shares are independent of the historical Keccak/NTT component shares.

Table XII reports whole-launch LSU operation counters and the stack watermark for one request. These counts are neither transferred bytes nor off-chip traffic, and they do not isolate register spills. The stage and whole-round arms have identical load/store counts; fusion primarily changes the permutation instruction sequence. The stage permutation loop itself contains no memory instructions, whereas the surrounding sponge and KEM still access memory.

TABLE XI
SINGLE-REQUEST ML-KEM PHASE INTERVALS, MILLIONS OF CYCLES.
REMAINDER = WHOLE LAUNCH MINUS PHASE SUM.

Backend	KeyGen	Encaps	Decaps	Remainder
SG1 C	9.838	11.038	13.318	0.495
SG25 shuffle	3.580	4.639	5.622	0.480
SG25 stages	2.789	3.830	4.649	0.482
SG25 whole round	2.722	3.762	4.567	0.482
Pointer PE	2.981	4.027	4.899	0.523

TABLE XII
SINGLE-REQUEST WHOLE-LAUNCH MEMORY-OPERATION COUNTS AND
STACK WATERMARKS. ALL BACKENDS USE A 19,232-BYTE PEAK
ALLOCATION ARENA.

Backend	Loads	Stores	Stack bytes
SG1 C	628,039	512,165	1,912
SG25 shuffle	521,343	189,538	2,184
SG25 stages	164,422	162,082	2,156
SG25 whole round	164,422	162,082	2,156
Pointer PE	152,797	136,038	1,420
Redundant outer KEM	3,994,787	3,929,398	1,868

The redundant outer-KEM control incurs 7,924,185 load/store operations, versus 326,504 for the stage leader mapping, approximately 24.3 times as many. At eight requests the redundant version takes 160.039 million cycles, slower than serial SG1. Correct output and a low instruction count alone therefore do not establish an efficient SIMT mapping.

F. Final NTT-Keccak integration

Table XIII repeats the five-backend comparison after fixing the final W32 NTT and cooperative arithmetic configuration. Every arm uses the same combined hardware build, serial FIPS-202 call graph, and FIPS 203 KAT coins. Each request executes 140 Keccak-f permutations, 15 forward NTTs, and nine inverse NTTs; all public keys, secret keys, ciphertexts, and shared secrets match byte-for-byte. Consequently, only the Keccak backend varies.

TABLE XIII
FINAL INTEGRATED ML-KEM-768 ON SIMX, $8w \times 32t$. CYCLES ARE
MILLIONS PER LAUNCH; GAINS USE SG1 AT THE SAME M .

Backend	$M = 1$	Gain	$M = 8$	Gain
SG1 C	27.739	1.000	44.825	1.000
SG25 shuffle	7.900	3.511	19.177	2.337
SG25 stages	5.410	5.128	12.247	3.660
SG25 whole round	5.198	5.336	11.947	3.752
Pointer PE	5.912	4.692	8.902	5.035

Source: pqc/results/keccak_ntt_unified_mlkem.csv.

Stages achieve $5.128 \times / 3.660 \times$ over SG1 at $M = 1/8$. Relative to the matched shuffle implementation, the instruction-only gains are $1.460 \times / 1.566 \times$. Whole-round fusion adds $1.041 \times / 1.025 \times$ over stages. The latency-throughput crossover remains: stages and the whole-round unit beat the pointer PE at $M = 1$ by $1.093 \times$ and $1.137 \times$, whereas the pointer PE leads them at $M = 8$ by $1.376 \times$ and $1.342 \times$.

TABLE XIV
MATCHED INTEGER-ONLY ML-KEM AT RV32 AND RV64 ON SIMX.
CYCLES ARE MILLIONS PER COMPLETE LAUNCH.

ISA	M1 instr.	M1 cycles	M8 instr.	M8 cycles
Stage RV32IM	436,262	5.410	3,490,096	12.170
Stage RV64IM	405,716	5.190	3,245,728	12.226
KROUND RV32IM	415,822	5.198	3,326,576	11.938
KROUND RV64IM	391,996	5.044	3,135,968	11.971

Source: pqc/results/rv32im_rv64im_keccak.csv.

Full XRT integration runs for stages and whole round reproduce the KAT and retired-instruction counts. Their whole-launch cycle gaps from SimX are 0.004% and 0.093%, respectively. These runs cover the AFU and processor RTL. Table XIII uses a common all-enabled core for the functional and cycle comparison. Section VI uses four separate builds that retain the same NTT and enable either no Keccak hardware, stages, whole round, or the pointer PE.

G. Matched integer-only RV32/RV64 portability

Table XIV recompiles Stage and KROUND at RV32IM and RV64IM while holding the source revision, W8T32 geometry, KAT input, cache configuration, NTT, and cooperative arithmetic fixed. F and D are disabled at both XLENs; ELF attributes and disassembly confirm soft-float IM binaries without floating-point or atomic instructions. SimX, RTL, and XRT reproduce every M1 KAT with exact retired-instruction agreement. All eight SimX requests also pass byte-for-byte.

RV64 reduces Stage instructions by 7.00% and KROUND instructions by 5.73%. Their M1 cycles fall by 4.07% and 2.98%, respectively, while M8 cycles rise by 0.46% and 0.27%. The saved Keccak issues are only part of the complete launch: NTT, sampling, encoding, memory traffic, and remaining arithmetic do not halve with the issue count. At M8, the higher cycles per retired instruction offset the smaller stream. The routed cost of the same integer-only NTT/Stage/KROUND configurations appears in Table XVII.

VI. IMPLEMENTATION COST

A. Final NTT architecture and selection

The final NTT maps one transform to one 32-lane warp. Coefficients remain in lane GPRs across the seven layers. The three lane-local layers use NTTMUL.K; the four cross-lane layers use CT/GS NTTBF operations at XOR distances 16, 8, 4, and 2. A half-width bank shares 16 signed 16×16 multipliers across the warp. CT retains latency/II 6/1, while GS and W32 NTTMUL.K use 7/2. Static routing implements the five legal encoded XOR distances; this kernel uses four of them.

Table XV records the matched architecture sweep that selected this design. Every row in that historical sweep includes the same pointer Keccak PE, so the first row is the no-NTT-ISE denominator for this table only. The registered full bank closes 250 MHz. Sharing the multipliers reduces its full-core LUT count by 4.14% while changing matched complete-KEM

TABLE XV

HISTORICAL MATCHED V80 NTT ARCHITECTURE SWEEP AT 250 MHZ. DELTAS USE THE NO-NTT-ISE ROW; ALL ROWS INCLUDE THE SAME POINTER KECCAK PE. WNS IS NS.

NTT hardware	LUT	Δ LUT	FF	Δ FF	DSP	WNS
No NTT ISE	334,917	–	265,187	–	160	0.000
Full bank	356,447	+21,530	272,430	+7,243	192	0.000
Half bank	341,680	+6,763	270,865	+5,678	176	0.000

Source: pqc/results/ntt_v80_ppa.csv.

TABLE XVI

MATCHED FULL-CORE V80 POST-ROUTE RESULTS AT 250 MHZ. DELTAS USE NTT ONLY AS THE DENOMINATOR. WNS IS NS.

Core	LUT	Δ LUT	FF	Δ FF	DSP	WNS
NTT only	330,530	–	267,838	–	176	0.000
NTT + stages	334,561	+4,031	269,806	+1,968	176	0.006
NTT + whole round	332,710	+2,180	274,237	+6,399	176	0.003
NTT + pointer PE	341,751	+11,221	271,058	+3,220	176	0.004

Source: pqc/results/ntt_keccak_backend_ppa.csv.

cycles by at most 0.041%. Relative to the historical no-NTT-ISE core, the selected half bank adds 2.02% LUTs, 2.14% FFs, and 16 DSPs without adding BRAM.

The full-bank-to-half-bank comparison holds instruction semantics and the kernel binary fixed, but the table predates the independently gated merged source used below. We therefore use it to justify the NTT architecture selection and do not subtract its absolute counts from the new Keccak rows.

B. Matched full-core Keccak backends

We target `xcv80-lsva4737-2MHP-e-S` with Vivado 2025.1, using post-route physical optimization. Table XVI compares four independent full-core builds under the same 250 MHz target. Each contains the selected NTT and arithmetic path. The NTT-only row is the common physical denominator; each other row enables exactly one Keccak hardware backend. The stage overhead is 4,031 LUT (1.22%) and 1,968 FF (0.73%). Whole round adds 2,180 LUT (0.66%) and 6,399 FF (2.39%), whereas the pointer PE adds 11,221 LUT (3.39%) and 3,220 FF (1.20%). BRAM and DSP counts are unchanged.

These independently gated builds supply the area denominator for the final ML-KEM comparison. They do not replace the common all-enabled binary used to measure cycles, whose purpose is to keep software and timing-model context identical. Post-route full-core differences include integration, placement, and routing effects in addition to the selected unit hierarchy. All four builds close 250 MHz, with WNS from 0.000 to 0.006 ns. The routed stage, whole-round, and pointer hierarchies contain 902/2,025, 2,503/6,130, and 10,414/4,556 LUT/FF, respectively. The whole-round critical path remains in issue-to-integer-ALU routing rather than inside the new unit.

C. Matched integer-only RV32/RV64 implementation

Table XVII holds the V80 device, W8T32 geometry, 250 MHz target, Vivado version, optimization level, and selected half-bank NTT fixed. F and D are disabled at both XLENs. It compares NTT-only, Stage, and KROUND builds; every row

TABLE XVII

MATCHED INTEGER-ONLY RV32/RV64 V80 POST-ROUTE RESULTS. BRAM IS TILE-EQUIVALENT, WNS IS NS, AND STA FREQUENCY IS MHZ UNDER A 250 MHZ TARGET.

Core	LUT	FF	BRAM	DSP	WNS	STA MHz
RV32 NTT only	220,950	131,798	133.0	112	0.004	250.3
RV32 NTT + stages	223,444	133,512	133.0	112	0.004	250.3
RV32 NTT + whole round	222,935	138,065	133.0	112	0.007	250.4
RV64 NTT only	429,067	228,266	261.5	16	0.000	250.0
RV64 NTT + stages	432,843	232,350	261.5	16	-0.148	241.1
RV64 NTT + whole round	431,004	234,240	261.5	16	-0.013	249.2

Source: pqc/results/rv32im_rv64im_keccak_ppa.csv.

TABLE XVIII

ISOLATED UNITS ON V80, 300 MHZ TARGET.

Unit	LUT	FF	STA $f_{\max}$ (MHz)
Pointer PE	3,223	1,607	368.4
Stages	2,821	2,101	323.7
Whole round	2,610	6,282	359.6

is an independent, non-incremental post-route run with zero routing errors. Negative WNS rows are reported as timing misses rather than closed implementations. RV64 Stage has WNS -0.148 ns at 241.1 MHz, while RV64 KROUND reaches -0.013 ns and 249.2 MHz.

Relative to RV32 NTT-only, the RV64 NTT-only full-core resource changes are 94.19% for LUTs, 73.19% for FFs, 96.62% for BRAM tiles, and -85.71% for DSPs. Both FPU hierarchies are absent, while the narrow NTT remains at 16 DSP. The DSP decrease comes from the general multiply/divide unit: the RV32 implementation uses 96 DSPs there, whereas the serial RV64 implementation uses 0.

Within RV32, Stage adds 2,494 LUT (1.13%) and 1,714 FF (1.30%); KROUND adds 1,985 LUT (0.90%) and 6,267 FF (4.76%). Within RV64, the corresponding increments are 3,776/4,084 LUT/FF for Stage and 1,937/5,974 LUT/FF for KROUND. None changes BRAM or DSP count.

The wide core itself must also be charged. The historical 300 MHz base configuration grid uses 42,553 LUTs at $8w \times 4t$, 148,827 at $8w \times 16t$, and 325,407 at $8w \times 32t$. Consequently, a small extension overhead relative to the chosen wide core is not the cost of widening a narrow GPU. The SG25 study fixes the wide core to isolate instruction granularity.

D. Isolated-unit FPGA and ASIC comparisons

Tables XVIII and XIX compare isolated units. The pointer unit performs a complete permutation, stages execute one selected RV32 stage instruction, and the whole-round unit executes one selected half of a round. Unit area is therefore not a same-operation latency comparison.

The whole-round unit uses 7.48% fewer isolated FPGA LUTs than stages, but 2.990 times the FFs. On the predictive ASAP7 library [10], it uses 2.123 times the total cell area and 2.989 times the sequential area. Its larger register footprint is visible after full-core FPGA integration, but LUT-only normalization largely hides it.

TABLE XIX

ISOLATED ASAP7 RVT, TT, 1.00 V, 25°C. YOSYS 0.57 / OPENSTA 2.7.0; MATCHED 500 MHz MAPPING CONSTRAINT. AREA IS μm^2 ; THESE ARE MAPPED, NOT ROUTED, ASIC RESULTS.

Unit	Total area	Sequential area	STA MHz
Pointer PE	1,723.254	470.642	2,187.2
Stages	1,832.123	612.943	1,882.9
Whole round	3,889.186	1,831.831	2,041.2

At common 250 MHz, the 64-chain RTL results and full-core LUT counts give a whole-round versus stage throughput/LUT ratio of

$$\frac{481.664}{117.693} \times \frac{334,561}{332,710} \simeq 4.115. \quad (10)$$

Thus the whole-round unit also wins this FPGA efficiency metric for pure permutation. It would be misleading to claim that the stage design wins all area-normalized metrics because it has fewer pipeline registers. Conversely, dividing the same pure-permutation gain by isolated ASIC area does not yield a measured full-ASIC application efficiency. We omit power/energy claims because no workload-activity-based power measurement is available.

VII. RELATED WORK AND INTERPRETATION

PQRV optimizes Keccak and polynomial arithmetic for several RISC-V ISA variants [9]. It motivates an assembly baseline rather than reliance on portable C alone. Bolat et al. study a custom SHA-3 instruction and its pipeline/storage integration in a RISC-V CPU [11]. These approaches show that instruction acceleration and baseline selection already have substantial prior work; the extra question here is how cooperating SIMT lane registers change the design boundary.

ML-Cube combines register-oriented polynomial computation with GPU execution and describes a 25-thread shuffle mapping for Keccak in Sec. 5.1.1 [5]. We use this mapping as prior art and as a software denominator. We do not compare its commercial-GPU timing numerically against a single FPGA soft core.

Listing 3 of NVIDIA’s published application describes `keccak1.sync.aligned.u64`, where lanes 0–24 hold 64-bit input/output words and a warp-collective invocation performs one round [6]. The text also discusses multiple-round forms. It supplies a concrete semantic predecessor, not measurements for our platform. Our comparator has explicitly implemented RV32 low/high and RV64 full-word destinations, literal rounds, a four-stage datapath, and single-vector writeback. Its timing and PPA numbers belong to this implementation and are not measurements of NVIDIA hardware.

The useful contribution is the controlled granularity study and its end-to-end consequences. Stages expose individually testable and replaceable collectives at lower sequential cost. Whole-round fusion provides the best measured GPR permutation throughput and competitive FPGA LUT efficiency. The pointer PE leads the saturated ML-KEM workload despite

architectural memory transfer. Which point to adopt depends on the target workload and the area metric, not solely on the number of custom instructions.

VIII. LIMITATIONS AND NEXT MEASUREMENTS

Historical component fractions apply to their fixed input, instrumented build, and narrow-core configuration. Direct interval profiling of the final $8w \times 32t$ implementation is still needed to separate absorb, extract, transforms, other arithmetic, wrappers, and allocation. The small stage-to-round ML-KEM gain does not identify these residual percentages.

The final integrated table lacks a matched $8w \times 32t$ PQRV assembly arm. Historical SG5 software uses a 16-thread configuration. A matched SG1/SG5/SG25 sweep and a loop-unrolling control would sharpen attribution; the historical experiments cannot substitute for these controls.

ML-DSA collective integration and mixed-input batches remain unmeasured. Existing ML-DSA results cover software and pointer controls on their own configurations. We report neither confidence intervals from repeated deterministic simulation nor quantiles from the incomplete host sweep.

Evidence includes XRT integration simulation and out-of-context full-core FPGA routing. Board execution, complete-system PPA, full-core routed ASIC results, and activity-driven energy remain outside the demonstrated scope. Each XLEN physical row is one implementation seed. Register residence alone does not establish side-channel resistance.

IX. CONCLUSION

The software baseline determines both which primitive to accelerate and which speedup is meaningful. On the historical ML-KEM profile, Keccak accounts for 65.65% and NTT/INTT for 12.67%; independent-request batching and optimized software provide additional, separately measured baselines.

On a fixed RV32 $8w \times 32t$ Vortex core, three subgroup stages convert the 25-word Keccak state into an explicit GPR collective interface. They improve permutation throughput by $12.91\times$ over shuffles and, with the final NTT path fixed, complete ML-KEM by $5.128\times/3.660\times$ over SG1 at one/eight requests. Whole-round fusion adds about fourfold permutation throughput but only 2–4% application speedup, while the pointer PE remains strongest in the eight-request launch. Integer-only RV64 halves the Keccak issue count, reduces complete-launch instructions by 5.7–7.0%, and slightly raises the eight-request cycle result. The result is an implemented set of design choices with measured tradeoffs, rather than a universal ranking inferred from one primitive benchmark.

APPENDIX A

DATA PROVENANCE AND REPRODUCTION

All local measurements quoted here are archived under `pqc/results/`. The companion README maps tables to source files. The generator recomputes ratios, checks ablation differences and phase sums, and emits vector figures and a SHA-256 input manifest. It uses data rows rather than historical narrative percentages. Collective datasets include

executable hashes, but their source tree has local changes: HEAD and input hashes cannot recover missing experiment source snapshots. Build with `make` in `pqc/docs/paper`; the portable source package also contains frozen generated tables and figures.

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